$y = mx + c$

$y_1 = m x_1 + c \quad -①$

$y_2 = m x_2 + c \quad -②$

(x_1, y_1) (x_2, y_2)

$y_1 - y_2 = m(x_1 - x_2)$

$m = \dfrac{y_1 - y_2}{x_1 - x_2} = \dfrac{y_2 - y_1}{x_2 - x_1}$

$y = ax^2 + bx + c$

$$\begin{bmatrix} 2 = a + b + c \\ 3 = 4a + 2b + c \\ 0 = 9a + 3b + c \end{bmatrix}$$

$(1, 2)$ $(2, 3)$ $(3, 0)$

$3a + b = 1$

$5a + b = -3$

$2a = -4, \quad a = -2$

$b = 7$

$2 = -2 + 7 + c$

$c = -3$

$$\boxed{y = -2x^2 + 7x - 3}$$

$$\begin{bmatrix} c & & & \\ & & & \\ c & & & \end{bmatrix} \rightarrow [\ \ ]$$

$$\begin{bmatrix} 2 & 1 & 1 & 1 \\ 3 & 4 & 2 & 1 \\ 0 & 9 & 3 & 1 \end{bmatrix}$$

$$[-2, 7, -3]$$

```
def solve(Eqns):
    # returns an array of solutions

>> Solve([[5, 2]])                    5 = 2x
[2.5]

>> Solve([[5, 2, 1], [7, 1, -1]]      5 = 2x + y
[4, -3]                               7 = x - y

>> Solve([[1, 1, -1, 1], [2, 1, 1, -1], [0, 1, -1, -1]])   1 = a - b + c
[1.5, 1, 0.5]                                              2 = a + b - c
                                                           0 = a - b - c
```

data

x	y
1	2
2	0

$y = mx + c$

$$\begin{bmatrix} 2 = m + c \\ 0 = 2m + c \end{bmatrix} \rightarrow \text{Solve} \rightarrow [\ ,\ ]$$

data

x	y
1	2
2	4
3	0

$y = ax^2 + bx + c$

$$\begin{bmatrix} 2 = a + b + c \\ 4 = 4a + 2b + c \\ 0 = 9a + 3b + c \end{bmatrix} \rightarrow \text{Solve}() \rightarrow [\ ,\ ,\ ]$$